

Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	RWASHANDE, Seith
Date of birth:	2/4/1990
Subject:	030-Flight Performance&Planing
Result:	29,3 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	384		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
2	391		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
3	382		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
4	379		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
5	420		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
6	373		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
7	388		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
8	381		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
9	377		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
10	315		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	375		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
12	374		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
13	380		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
14	361		31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
15	278	31.3.3	Basic calculations of CG	0,00	2,00
16	423	32.	PERFORMANCE - AEROPLANES	1,00	1,00
17	404	32.	PERFORMANCE - AEROPLANES	0,00	1,00
18	328	32.	PERFORMANCE - AEROPLANES	1,00	1,00
19	231	32.2.5.4	Landing	0,00	1,00
20	179	32.3.3.1	Take-off	0,00	1,00
21	31	32.4.1	Take - off	0,00	1,00
22	115	32.4.1	Take - off	0,00	1,00
23	34	32.4.1.7	Take-off climb	1,00	1,00
24	197	32.4.1.7	Take-off climb	0,00	1,00
25	12	32.4.1.7	Take-off climb	0,00	1,00
26	119	32.4.1.8	Obstacle-limited take-off	1,00	1,00
27	29	32.4.1.8	Obstacle-limited take-off	0,00	1,00
28	133	32.4.3	Cruise	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	51	32.4.3	Cruise	1,00	1,00
30	154	32.4.3.4	Long Range Cruise	1,00	1,00
31	18	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
32	421	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
33	417	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	0,00	1,00
34	221	33.1.1.1	Routes, airfields, heights and altitudes from VFR charts	1,00	1,00
35	2	33.2.1	IFR Navigation plan	0,00	1,00
36	175	33.2.1.2	Courses and distances from en-route charts	0,00	1,00
37	122	33.3.3	Specific fuel calculation procedures	0,00	1,00
38	252	33.4.1.3	Airway routings and airspace structure	0,00	1,00
39	224	33.4.2.1	Extraction and analysis of relevant data from meteorological documents	1,00	1,00
40	203	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:29%				12,00	41,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: RWASHANDE, Seith

Location: Uganda Civil Aviation Authority

Your Result: 12,00 from 41,00 Points (29,27 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	4,00	16,00	25,00
31.3. FUNDAMENTALS OF CG CALCULATIONS	1	0,00	2,00	0,00
32. PERFORMANCE - AEROPLANES	15	6,00	15,00	40,00
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	1	0,00	1,00	0,00
32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES	1	0,00	1,00	0,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	10	4,00	10,00	40,00
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	2,00	10,00	20,00
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	2	1,00	2,00	50,00
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	2	0,00	2,00	0,00
33.3. FUEL PLANNING	1	0,00	1,00	0,00
33.4. PRE-FLIGHT PREPARATION	2	1,00	2,00	50,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	1	0,00	1,00	0,00
Total	40	12,00	41,00	29,27

1

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☐ The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.
- ☒ Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.
- ☐ The aft CG provides the highest stall speed and cruising speed.

2

As the CG location is changed, recovery from a stall becomes progressively

☐ more difficult as the CG moves either forward or rearward.

☐ more difficult as the CG moves rearward.

☒ less difficult as the CG moves rearward.

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3

The center of gravity of an aircraft can be determined by

- ☐ dividing total arm by total moments.
- ☐ multiplying total arm by total weight.
- ☒ dividing total moment by total weight.

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4

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☐ 65.8 pounds.

☐ 62.5 pounds.

☒ 68.9 pounds.

5

The center of gravity tube is

☒ lengthened for heavier pilots.

☐ lengthened for lighter pilots.

☐ shortened for lighter pilots.

6

How much weight could be added at Station 160 without exceeding the aft CG limit?

Aircraft weight 8,300 lb
CG location Station 90.0
Aft CG limit Station 90.5

☐ 16.5 pounds.

☒ 59.7 pounds.

☐ 13.9 pounds.

7

What is characteristic of the indicated airspeed if the CG is at the most forward allowable position and constant power and altitude are maintained?

- ☐ Indicated airspeed will be less than it would be with the CG in the most rearward allowable position.
- ☐ There is no relationship between CG location and indicated airspeed.
- ☒ Indicated airspeed will be greater than it would be with the CG in the most rearward allowable position.

8

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☒ Yes; the new CG would be located at Station 117.69.

☐ No; the new CG would be located at Station 118.15.

☐ Yes; the CG would remain at Station 115.8.

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9

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 83.4.

☐ 80.6.

☒ 81.3.

10

The term 'weigh-off' as used in ballooning means to determine the

- ☒ standard weight and balance of the balloon.
- ☐ amount of fuel required for ascent to a preselected altitude.
- ☐ static equilibrium of the balloon as loaded for flight.

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11

What is the maximum weight that could be added at Station 150.0 without exceeding the aft CG limit?

Aircraft weight 5,000 lb
CG location Station 80.0

☐ 70.0 pounds.

☒ 35.9 pounds.

☐ 69.5 pounds.

12

How much weight could be added at Station 120 without exceeding the aft CG limit?

Aircraft weight 9,500 lb

CG location Station 90.0

Aft CG limit Station 90.5

☐ 110.5 pounds.

☐ 161.0 pounds.

☒ 61.0 pounds.

13

Could 100 pounds of weight be shifted from Station 130.0 to Station 30.0 without exceeding the forward CG limit?

Total weight 2,800 lb
CG location Station 120.0

- ☐ No; the new CG would be located at Station 116.89.
- ☒ Yes; the new CG would be located at Station 117.89.
- ☐ No; the new CG would be located at Station 116.42.

14

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☒ 953.6 pounds.

☐ 968.6 pounds.

☐ 924.5 pounds.

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15

(Please view annex 031-038 rev. 05.11.2004)
For the following boom to be in balance:

☐ $A = B \times Fa / Fb$

☐ $A = B + Fb / Fa$

☐ $A = B \times Fb / Fa$

☒ $A = B (Fa + Fb)$

See Attachments:

1. 031-038.gif

Annex

(Attachment
No.1)

16

To level off at an airspeed higher than the descent speed, the addition of power should be made, assuming a 500 FPM rate of descent, at approximately

- ☐ 50 to 100 feet above the desired altitude.
- ☐ 150 to 200 feet above the desired altitude.
- ☒ 100 to 150 feet above the desired altitude.

17

The best lift/drag ratio of a glider occurs when parasite drag is

- ☐ equal to induced drag.
- ☒ less than induced drag.
- ☐ equal to total drag.

18

(Refer to Figures 84 and 85.) What is the approximate fuel consumed when holding under Operating Conditions H-4?

☐ 3,050 pounds.

☒ 3,190 pounds.

☐ 2,550 pounds.

See Attachments:

1. atp_84.jpg	figure 84	(Attachment No.2)
2. atp_85.jpg	figure 85	(Attachment No.3)

19

To minimize the risk of hydroplaning during landing the pilot should:

- ☒ postpone the landing until the risk of hydroplaning no longer exists.
- ☐ make a "positive" landing and apply maximum reverse thrust and brakes as quickly as possible.
- ☐ use normal landing-, braking- and reverse technique.
- ☐ use maximum reverse thrust, and should start braking below the hydroplaning speed.

20

The decision speed at take-off (V1) is the calibrated airspeed:

- ☒ below which the take-off must be continued.
- ☐ below which take-off must be rejected if an engine failure is recognized, above which take-off must be continued.
- ☐ at which the failure of the critical engine is expected to occur.
- ☐ at which the take-off must be rejected.

21

How is V2 affected if take off flaps 20° is chosen instead of take off flaps 10°?

- ☐ V2 has the same value in both cases.
- ☐ V2 increases in proportion to the angle at which the flaps are set.
- ☒ V2 decreases if not restricted by VMCA.
- ☒ V2 has no connection with T/O flap setting, as it is a function of runway length only.

22

When the outside air temperature increases, then

- ☐ the field length limited take-off mass and the climb limited take-off mass decreases.
- ☒ the field length limited take-off mass increases but the climb limited take-off mass decreases.
- ☐ the field length limited take-off mass decreases but the climb limited take-off mass increases.
- ☐ the field length limited take-off mass and the climb limited take-off mass increases.

23

Which of the following statements is correct?

- ☐ The take-off distance with one engine out is independant of the wind component.
- ☐ The accelerate stop distance required is independant of the runway condition.
- ☐ The performance limited take-off mass is independant of the wind component.
- ☒ The climb limited take-off mass is independant of the wind component.

24

The best rate of climb at a constant gross mass

- ☒ increases with increasing altitude due to the higher true airspeed.
- ☐ increases with increasing altitude since the drag decreases due to the lower air density.
- ☐ decreases with increasing altitude since the thrust available decreases due to the lower air density.
- ☐ is independent of altitude.

25

The rate of climb

- ☐ is angle of climb times true airspeed.
- ☐ is the downhill component of the true airspeed.
- ☒ is approximately climb gradient times true airspeed divided by 100.
- ☒ is the horizontal component of the true airspeed.

26

Regarding the obstacle limited take-off mass, which of the following statements is correct?

- ☐ The maximum bank angle which can be used is 10°.
- ☐ Wind speed plays no role when calculating this particular mass.
- ☐ The obstacle limited mass can never be lower than the climb limited take-off mass.
- ☒ A take-off in the direction of an obstacle is also permitted in tail wind condition.

27

Which statement, in relation to the climb limited take-off mass of a jet aeroplane, is correct?

- ☒ 50% of a head wind is taken into account when determining the climb limited take-off mass.
- ☐ The climb limited take-off mass decreases with increasing OAT.
- ☐ The climb limited take-off mass is determined at the speed for best rate of climb.
- ☐ On high elevation airports equipped with long runways the aeroplane will always be climb limited.

28

With regard to a unaccelerated horizontal flight, which of the following statement is correct?

- ☐ The minimum drag is independant of the aircraft mass.
- ☒ The minimum drag is a function of the pressure altitude.
- ☐ The minimum drag is is a function of the density altitude.
- ☐ The minimum drag is proportional to the aircraft mass.

29

Which statement with respect to the step climb is correct?

- ☐ A step climb is executed because ATC desires a higher altitude.
- ☒ Executing a desired step climb at high altitude can be limited by buffet onset at g-loads larger than 1.
- ☐ A step climb must be executed immediately after the aeroplane has exceeded the optimum altitude.
- ☐ A step climb is executed in principle when, just after leveling off, the 1.3g altitude is reached.

30

A lower airspeed at constant mass and altitude requires

- ☒ **a higher coefficient of lift.**
- ☐ **less thrust and a lower coefficient of lift.**
- ☐ **more thrust and a lower coefficient of drag.**
- ☐ **more thrust and a lower coefficient of lift.**

31

Which of the following statements regarding filing a flight plan is correct?

- ☐ A flying college can file repetitive flight plan for VFR flights.
- ☐ A flight plan should be filed when a national FIR boundary will be crossed.
- ☒ Any flight plan should be filed at least 10 minutes before departure.
- ☐ A flight plan should be filled atleast one hour before departure

32

How is your flight plan closed when your destination airport has IFR conditions and there is no control tower or Flight control Center(FCC) on the field?

- ☐ Upon landing, you must close your flight plan by radio or by telephone to any FCC or ATC facility.
- ☒ You may close your flight plan any time after starting the approach by contacting any FCC or ATC facility.
- ☐ The ARTCC controller will close your flight plan when you report the runway in sight.

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33

How long will a Flight Service Station hold a VFR flight plan past the proposed departure time?

☒ **30 minutes.**

☐ **2 hours.**

☐ **1 hour.**

34

VFR flights shall not be flown over the congested areas of cities at a height less than

- ☐ D the heighest obstacle.
- ☒ A 1000 ft above the heighest obstacle within a radius of 600 m from the aircraft.
- ☐ C 500 ft above the heighest obstacle.
- ☐ B 2000 ft above the heighest obstacle within a radius of 600 ft from the aircraft.

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35

In an ATS flight plan, Item 15 (route), a cruising pressure altitude of 32000 feet would be entered as :

☐ C 32000

☐ B S3200

☒ A FL320

☐ D F320

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36

You are flying a constant compass heading of 252° . Variation is 22°E , deviation is 3°W and your INS is showing a drift of 9° right. True track is ?

☐ 280°

☒ 242°

☐ 224°

☐ 262°

37

An aircraft flies at an airspeed of 380 kt. It flies from A to B and back to A. Distance AB = 480 NM. When going from A to B, it experiences a headwind component = 60 kt. The wind remains constant. The duration of the flight will be:

☒ 3h 00min

☐ 2h 35min

☐ 2h 32min

☐ 2h 10min

38

From which of the following would you expect to find the dates and times when temporary danger areas are active

☒ **Only AIP**

☐ **RAD/NAV charts**

☐ **NOTAM and AIP**

☐ **SIGMET**

39

A METAR reads : SA1430 35002KT 7000 SKC 21/03 Q1024 =
Which of the following information is contained in this METAR?

☒ *temperature/dewpoint*

☐ *day/month*

☐ *period of validity*

☐ *runway in use*

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40

When completing an ATC flight plan, an elapsed time (Item 16) of 1 hour 55 minutes should be entered as:

☐ 115M

☒ 1H55

☐ 0155

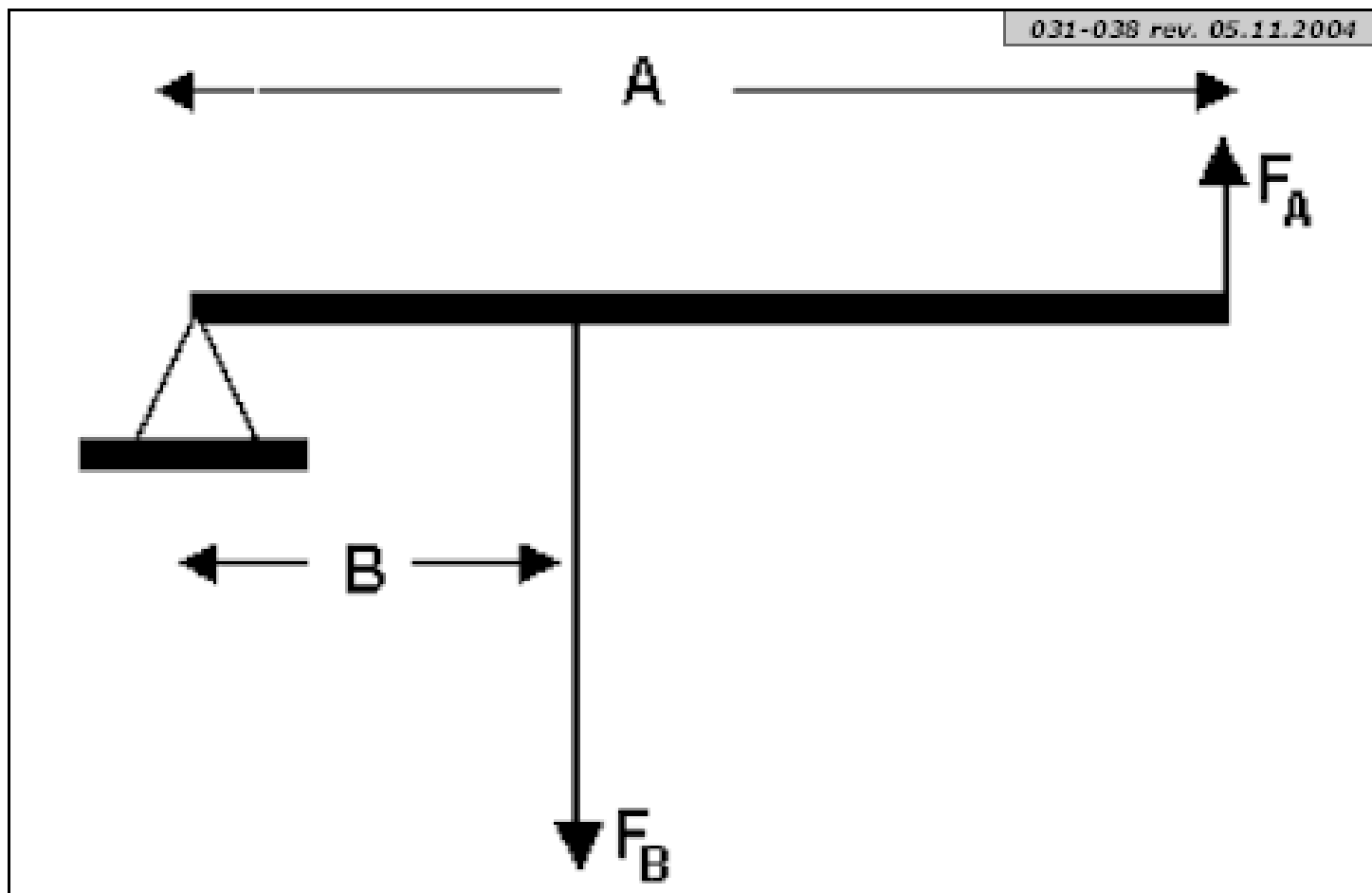
☐ 0115

Attachment No. 1

Questions 15

Annex

031-038.gif



Attachment No. 2

Questions 18

figure 84

atp_84.jpg

OPERATING CONDITIONS	H-1	H-2	H-3	H-4	H-5
ALTITUDE	24,000	17,000	8,000	18,000	22,000
WEIGHT (X1000)	195	185	155	135	175
ENGINES OPERATING	3	3	3	3	3
HOLDING TIME (MIN)	15	30	45	25	35

Figure 84. B-727 – Holding © ASA

Attachment No. 3

Questions 18

figure 85

atp_85.jpg

EPR IAS - KTS FF PER ENG - LB/HR			HOLDING							B-727	
PRESSURE ALTITUDE FT	GROSS WEIGHT - 1000 LB										
	200	190	180	170	160	150	140	130	120		
25000	1.85 268 3600	1.81 261 3400	1.77 253 3210	1.73 246 3030	1.69 238 2860	1.64 230 2680	1.60 222 2510	1.55 213 2340	1.51 205 2180		
20000	1.69 265 3630	1.66 258 3450	1.62 251 3280	1.59 244 3110	1.55 236 2940	1.51 228 2770	1.48 220 2600	1.44 212 2440	1.40 204 2270		
15000	1.56 263 3670	1.53 256 3500	1.50 249 3340	1.47 242 3170	1.44 235 3000	1.41 227 2850	1.38 219 2680	1.35 211 2520	1.32 203 2350		
10000	1.45 262 3800	1.43 255 3640	1.40 248 3460	1.38 241 3310	1.35 234 3140	1.33 226 2970	1.30 218 2810	1.28 210 2640	1.25 202 2480		
5000	1.36 260 3890	1.34 254 3720	1.32 247 3550	1.30 240 3380	1.28 233 3220	1.26 225 3060	1.24 218 2890	1.22 210 2730	1.20 201 2560		

Figure 85. B-727 – Holding Performance Chart © ASA